Sohan Jayram Reddy Kadiri

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PROFESSIONAL SUMMARY

AI Software Engineer with experience building LLM-powered applications, RAG pipelines and AI infrastructure using Python, Go, TypeScript and AWS. Skilled in model integration, vector search, inference optimization and automated evaluation.

EDUCATION

Northeastern University Sep 2025 - May 2027
Master of Science, *Computer Science* GPA: 3.9

Siddaganga Institute of Technology Dec 2021 - Jun 2025
Bachelor of Technology, *Artificial Intelligence and Data Science*

WORK EXPERIENCE

Siemens | Software Engineer Intern | Remote Aug 2024 - Jul 2025

- Engineered 2 **AI-powered full-stack applications using React, Node.js and PostgreSQL**, serving 2,500+ users with 250 ms p95 API latency.
- Improved chatbot retrieval relevance by 40% through a **Pinecone-backed RAG pipeline**, supporting 1,200+ conversations with 850 ms p95 response latency.
- **Led 3 engineering interns** through AI feature development and REST API integration, resolving 9 production issues through debugging, log analysis and service-level testing.
- Automated **Docker-based CI/CD pipelines** using Azure and Vercel, reducing deployment time by 70%, from 20 minutes to 6 minutes.

Unisys | Software Engineering Intern | Remote Apr 2024 - Jun 2025

- Developed **RESTful backend services** for an e-commerce marketplace serving 8,000+ users and processing 1,600+ orders, generating $20K in gross merchandise value.
- Optimized **checkout APIs and transaction workflows** through one-click checkout and funnel improvements, increasing conversion by 22% and contributing $14K in revenue.
- Engineered **real-time messaging using WebSockets**, supporting 350 concurrent connections with sub-150 ms delivery latency and 99.8% message delivery.
- Improved mobile conversion by 8% through **Lighthouse-guided performance optimizations**, reducing page load time to under 1.2 seconds.

PROJECTS

Harbor — LLM Inference & Reliability Platform | Go, Python, FastAPI, Redis, PostgreSQL, pgvector, Docker, Prometheus

- Engineered an **OpenAI-compatible inference gateway in Go** with streaming and model routing, integrating a RAG reference application backed by PostgreSQL and pgvector.
- Implemented **embedding-based semantic caching, Redis rate limiting and provider failover**, achieving an 85% cache-hit rate in a 400-request simulated benchmark.
- Accelerated cached responses by **9.5×** through semantic caching, reducing median latency from approximately 1,155 to 120 ms against a mock inference provider.

Lumen — Distributed AI Evaluation & Reliability Platform | Python, FastAPI, Celery, Redis, PostgreSQL, React, Prometheus, Kubernetes

- Engineered a **distributed LLM evaluation platform using FastAPI, Celery and Redis**, processing 10,000 evaluation cases per run through asynchronous workers.
- Implemented **five LLM evaluation methods**, including embedding similarity and LLM-as-judge, using the Mann–Whitney U test to detect statistically significant quality regressions.
- Integrated **Prometheus, Grafana and automated quality alerts**, achieving 95% regression-detection recall across 200 labeled test cases.

InspectIQ — AI-Powered Vehicle Inspection Platform | Python, TypeScript, React, AWS Bedrock, Lambda, SQS, S3, PostgreSQL

- Developed an **AI-assisted vehicle inspection platform using Amazon Bedrock**, integrating image analysis, OCR and damage identification to generate structured inspection findings.
- Built an **asynchronous S3–SQS–Lambda inference pipeline** with schema validation, retries and human approval before AI findings enter inspection reports.
- Implemented **AI-output validation and regression testing** using a 108-image challenge corpus, achieving 95% schema-valid outputs and reducing review time by 30%.

SKILLS

Languages: Python, Go, TypeScript, Java, C++, JavaScript, SQL
AI & ML: LLMs, RAG, Prompt Engineering, Embeddings, Semantic Search, Model Evaluation, PyTorch, TensorFlow, Scikit-learn
AI Frameworks: FastAPI, Amazon Bedrock, Pinecone, pgvector, Sentence Transformers, OpenAI-compatible APIs
Backend & Data: Node.js, PostgreSQL, Redis, MongoDB, Kafka, Celery, REST APIs
Cloud & MLOps: AWS (EC2, S3, Lambda), Azure, Docker, Kubernetes, Git, GitHub Actions, CI/CD, Prometheus, Grafana

CERTIFICATIONS & ACHIEVEMENTS

- Open Source - Merged PR #472 into mindsdb/anton, fixing a bug that deleted saved user rules and adding 3 regression tests. Sep 2026
- Publication - "Detection of DDoS Attacks in Fog Computing using Interpretable Quantum-based 2Q Learning Approach" - IEEE Jan 2025